

SECJ1023 PRACTICAL PAPER - PAST QUESTION SAMPLE

Question 1

[40 marks]

Given a C++ program named **program1.cpp** with four classes named, `Owner`, `Car`, `CarForSale` and `CarForRent`. Based on the program, answer the following questions. Write your answer for all questions below in the attached **docx file**. Alternatively, you can write your answers on papers and take snapshots of them with your smartphone.

- a. Draw the class diagram based on the implementation done in the program.

(24 marks)

- b. Determine whether the following classes have any relationship to one another in the context of Object-Oriented Programming (OOP). If so, specify the type of relationship, otherwise, write No Relationship.
 - i. `Car` and `Owner`
 - ii. `CarForRent` and `Owner`
 - iii. `CarForSale` and `Car`

(6 marks)

- c. Determine whether the following methods can be accomplished. If so, complete the code of the method. Otherwise, explain why it is impossible. Note that the goal of each method is as the name implies.
 - i. `CarForSale::getOwnerName()`
 - ii. `CarForRent::getOwnerMobileNumber()`

(10 marks)

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Question 2

[60 marks]

Given **program2.cpp** that will ask the user to choose a three-dimensional object either cuboid or sphere, then enter the size of information about the object accordingly, and finally print out the volume of the object. If the user enters other inputs, the program will display an error message "Invalid input". Figure 1 shows some example runs of the program. Note that the **bold** texts indicate user inputs. **Note:** As for all questions (a) until (e), rewrite the codebase program into the same file and submit as **program2.cpp**.

Run 1

```
Enter the type of object 1: Cuboid, 2: Sphere => 1
Enter the cuboid's dimensions, width, length and height => 2 3 4

Cuboid dimensions: 2 x 3 x 4
Volume: 24
```

Run 2

```
Enter the type of object 1: Cuboid, 2: Sphere => 2
Enter the radius => 4.5

Sphere's radius: 4.5
Volume: 381.51
```

Run 3

```
Enter the type of object 1: Cuboid, 2: Sphere => 5

Invalid input
```

Figure 1: Example results of the program

Rewrite the program using the polymorphism concept and handling the user input errors with exceptions. The new program should produce the same result as the original program. You are required to add two more classes to your program. You may use the inline style to define all the classes. Below are the expected elements your program should include:

- Definition of the class `ThreeDimensionalObject`. (10 marks)
- Definition of the class that represents cuboid objects. (15 marks)

- c. Definition of the class that represents sphere objects. (15 marks)
- d. Definition for the main function. (14 marks)
- e. Handling the user input error using an exception approach (6 marks)